TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
ADVANCED COLLEGE OF ENGINEERING AND MANAGEMENT
DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING
KALANKI, KATHMANDU

A Major Project Final Report On

AgroLedger: A Blockchain-Based Agricultural Supply Chain Management System

[CT-755]

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A Major Project Final report submitted to the Department of Electronics and Computer Engineering in the partial fulfillment of the requirements for degree of Bachelor of Engineering in Computer/Electronics Engineering
Kathmandu, Nepal

31st March, 2026

ADVANCED COLLEGE OF ENGINEERING AND MANAGEMENT

DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

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ACKNOWLEDGEMENT

We take this opportunity to express our deepest and sincere gratitude to our Project Supervisor **Er. Hemant Joshi**, Department of Electronics and Computer Engineering for his insightful advice, motivating suggestions, invaluable guidance, help, and support in the successful completion of this project, and also for his/her constant encouragement and advice throughout our Bachelor's program.

We express our deep gratitude to **Er. Roshani Ghimire**, Head of the Department of Electronics and Computer Engineering, **Dr. Raksha Dangol** and **Er.Navaraj Banstola**, Deputy Head, Department Of Electronics and Computer Engineering, for their regular support, cooperation, and coordination.

The in-time facilities provided by the department throughout the Bachelor's program are also equally acknowledgeable.

We would like to convey our thanks to the teaching and non-teaching staff of the Department of Electronics and Communication and Computer Engineering, ACEM for their invaluable help and support throughout the period of Bachelor's Degree. We are also grateful to all our classmates for their help, encouragement, and invaluable suggestions.

Finally, yet more importantly, we would like to express our deep appreciation to our grandparents, parents, and siblings for their perpetual support and encouragement throughout the Bachelor's degree period.

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ABSTRACT

Agriculture plays a vital role in the economy of Nepal, however traditional agricultural supply chains suffer from issues such as lack of transparency, poor traceability, and inefficient record-keeping, where information is often manually maintained or stored in centralized systems, making it difficult to verify product authenticity and ensure reliable data sharing among stakeholders. To address these challenges, this project proposes AgroLedger, a blockchain-based agricultural supply chain management system that leverages Ethereum blockchain technology and smart contracts to create a decentralized and tamper-proof platform for recording transactions and tracking product movement. The system enables end-to-end traceability, allowing stakeholders and consumers to view the complete journey of agricultural products from origin to destination, thereby improving trust and accountability. During the development process, several challenges were encountered, including understanding blockchain architecture, implementing smart contracts, integrating the frontend with blockchain networks, and handling transaction delays and gas-related issues; these challenges were addressed using tools such as Truffle for development, Ganache for local blockchain testing, and MetaMask for secure wallet-based authentication and transaction management. The final implementation demonstrates that blockchain technology can effectively enhance transparency, ensure data integrity, and improve coordination among stakeholders, providing a practical solution for modernizing agricultural supply chains in developing countries like Nepal.

Keywords: *Agriculture, Blockchain, DApp, Ethereum, Smart Contracts, Supply Chain Management, Traceability*

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ABBREVIATIONS

DFD: Data Flow Diagram
ETH: Ethereum
IoT: Internet of Things
KYC: Know Your Customer
PoA: Proof of Authority
PoS: Proof of Stake
SC: Smart Contract
SCM: Supply Chain Management
Tx: Transaction
UI: User Interface
UX: User Experience

CHAPTER 1
INTRODUCTION

1.1 Background

Agriculture is a fundamental sector in Nepal, supporting a large portion of the population and contributing significantly to the national economy. Traditionally, agricultural supply chains in Nepal have relied on manual processes and fragmented record-keeping systems, where information about production, storage, and distribution is either poorly documented or not shared efficiently among stakeholders. This lack of digital integration creates gaps in communication and coordination, making the overall system inefficient and difficult to manage.

As supply chains become more complex, these traditional approaches are no longer sufficient. Issues such as lack of transparency, poor traceability, data inconsistency, and limited accountability have become increasingly evident. Consumers often have no reliable way to verify the origin or quality of agricultural products, while stakeholders face challenges in tracking product movement across different stages. Studies have shown that traditional supply chain systems are prone to inefficiencies and fraud due to their centralized and opaque nature [1]. Furthermore, the absence of a unified and trustworthy system leads to reduced trust among participants and limits the potential for modernization.

With the advancement of digital technologies, blockchain has emerged as a promising solution for addressing these challenges. Blockchain provides a decentralized and immutable ledger that enables secure and transparent recording of transactions across multiple participants. Research highlights that blockchain can significantly enhance traceability, improve data integrity, and reduce inefficiencies in agricultural supply chains [2]. By transitioning from traditional methods to a blockchain-based approach, it becomes possible to establish a more transparent, reliable, and efficient system, which forms the foundation of this project.

1.2 Motivation

While studying the agricultural supply chain in the context of Nepal, we observed that the primary challenge is not limited to a single issue but rather the overall lack of a transparent and reliable system that connects all stakeholders. The existing system does not provide a clear and verifiable flow of information, making it difficult to ensure accountability, trace product history, and maintain trust among participants.

At the same time, there is a growing demand for food safety, product authenticity, and efficient supply chain management. Consumers are becoming more conscious about the quality and origin of agricultural products, while stakeholders require better coordination and data sharing mechanisms to improve efficiency. Traditional systems fail to meet these expectations due to their dependence on manual processes and centralized control.

This motivated us to explore the use of blockchain technology as a modern solution. Its decentralized nature, combined with features such as immutability and traceability, makes it suitable for transforming agricultural supply chains. Therefore, the motivation behind this project is to develop a blockchain-based system that improves transparency, enables end-to-end traceability, and enhances trust among all participants in the agricultural ecosystem.

1.3 Statement of Problems

The agricultural supply chain in Nepal continues to operate largely on traditional and fragmented systems, where information related to production, transportation, and distribution is not consistently recorded or shared among stakeholders. This lack of a unified and transparent system creates significant challenges in tracking the movement of agricultural products across different stages of the supply chain.

Due to the absence of reliable and tamper-proof data, stakeholders face difficulties in verifying product authenticity, ensuring quality, and maintaining accountability. Consumers have limited access to trustworthy information regarding the origin and handling of products, which raises concerns about food safety and reliability. At the same time, inefficient coordination and poor data visibility among stakeholders lead to delays, errors, and reduced overall system efficiency.

Existing centralized systems are often vulnerable to data manipulation, loss, and lack of trust, making them inadequate for managing modern agricultural supply chains. Therefore, there is a need for a secure, transparent, and decentralized system that enables accurate data sharing, ensures traceability, and improves trust and coordination among all participants in the agricultural ecosystem.

1.4 Project Objectives

a) To design and implement a decentralized system using blockchain technology for secure and transparent data management in the agricultural supply chain.

b) To enable end-to-end traceability of agricultural products, allowing stakeholders

and consumers to track product movement from origin to destination.

c) To ensure data integrity and security by using smart contracts for recording and managing transactions in a tamper-proof manner.

d) To improve coordination and trust among stakeholders by providing a unified platform for information sharing and verification.

1.5 Significance of Study

This study is significant as it addresses key challenges in the agricultural supply chain of Nepal, particularly the lack of transparency, traceability, and reliable information sharing among stakeholders. By introducing a blockchain-based system, the project demonstrates how modern technology can be utilized to improve the efficiency and trustworthiness of traditional agricultural practices.

The proposed system benefits farmers by providing better visibility of product movement and ensuring more reliable transaction records. It also benefits consumers by enabling them to verify the origin, quality, and handling of agricultural products, thereby increasing confidence in food safety. Additionally, other stakeholders such as distributors and retailers can benefit from improved coordination and access to accurate, tamper-proof data.

From an academic perspective, this study contributes to the growing field of blockchain applications in agriculture and supply chain management. It serves as a practical example of how decentralized technologies can be implemented to solve real-world problems, particularly in developing countries like Nepal. Furthermore, the system can be extended and improved in the future by integrating advanced technologies such as IoT and real-time data systems, making it a foundation for further research and development.

CHAPTER 2
LITERATURE REVIEW

Zheng et al. [3] studied how blockchain-based traceability systems can be adopted in agricultural supply chains using an evolutionary game theory approach. Their research focuses on understanding how different stakeholders such as farmers, cooperatives, and regulators make decisions over time. The study shows that farmers are often hesitant to adopt new technologies due to initial costs and uncertainty. However, when external factors such as government subsidies, strict regulations, and increasing consumer awareness are introduced, the likelihood of adoption increases significantly. One interesting insight from this work is that collaboration among stakeholders plays a key role in successful implementation. Although the study provides strong theoretical backing, it does not present a working system or prototype, which limits its practical application in real-world agricultural environments.

Tripoli and Schmidhuber [4] explored the broader opportunities of blockchain technology in the agri-food sector. Their work highlights how blockchain can help solve major issues such as food fraud, lack of transparency, and inefficient data sharing. The authors explain that blockchain allows all stakeholders to access a shared and tamper-proof record of transactions, which increases trust across the supply chain. They also emphasize that blockchain can support fair trade practices by ensuring that farmers receive accurate compensation. However, the study remains mostly conceptual and does not provide a detailed technical framework or implementation. As a result, while it offers valuable insights into the potential of blockchain, it does not directly guide how such systems can be developed or deployed.

Kamilaris et al. [1] provided a detailed survey of blockchain applications in agriculture and food supply chains. Their research categorizes different use cases, including product traceability, certification, and logistics management. The study explains how blockchain technology can create a transparent and immutable record of transactions, which helps reduce fraud and improve trust among stakeholders. Another important contribution is the discussion of technical aspects such as decentralization and consensus mechanisms, which make blockchain suitable for supply chain systems. However, the paper mainly reviews existing work and does not include experimental validation or system implementation. This makes it useful for understanding the field but less helpful for practical development.

Tian [2] proposed a blockchain-based traceability system combined with RFID technology to improve food safety. In this system, RFID tags are used to collect real-time data about agricultural products, which is then stored securely on the blockchain. This

approach ensures that product information cannot be altered, making the supply chain more transparent and reliable. The study demonstrates how such a system can help track products from farm to consumer, reducing the chances of fraud and contamination. However, the research does not address challenges such as scalability and cost, which are important factors when implementing such systems in developing countries like Nepal.

Kumarathunga et al. [5] focused on using blockchain technology in agricultural markets to ensure fairness and transparency. Their system uses smart contracts to automate agreements between farmers and buyers, which helps eliminate intermediaries and ensures fair pricing. This approach reduces the risk of exploitation and provides farmers with better control over their products. The study also highlights how blockchain can improve trust in market transactions. However, it mainly focuses on the financial and transactional aspects of agriculture and does not cover the complete supply chain process, such as transportation and storage.

Francisco and Swanson [6] examined the role of blockchain technology in improving supply chain management. Their research shows that blockchain can significantly enhance transparency by allowing all stakeholders to access real-time information about product movement. The study also highlights how blockchain can reduce delays and inefficiencies by eliminating the need for intermediaries. One of the key strengths of this work is its focus on real-world applications, such as IBM Food Trust. However, the solutions discussed are mainly designed for large-scale industries, making them less suitable for small-scale farmers in developing countries.

Caro et al. [7] proposed a blockchain-based traceability system that integrates Internet of Things (IoT) devices to enable realtime data collection within agricultural supply chains. Their approach focuses on capturing environmental and logistical data, such as temperature, humidity, and transportation conditions, and securely storing this information on a blockchain ledger. This ensures that the data remains immutable and can be accessed by all stakeholders, thereby improving transparency and trust across the supply chain. The integration of IoT and blockchain provides a comprehensive monitoring solution that enhances decision-making and reduces the risk of data manipulation. However, the implementation of such a system requires significant infrastructure investment, including IoT sensors and network connectivity, which may not be practical for small-scale farmers in developing countries.

Saberi et al. [8] conducted an extensive study on the application of blockchain technology in sustainable supply chain management. Their research highlights how blockchain can enhance security, improve transparency, and reduce operational inefficiencies by eliminating intermediaries and enabling direct interactions among stakeholders. The

study also emphasizes the environmental benefits of blockchain, such as reducing waste through improved tracking and resource management. Additionally, blockchain enables better auditing and compliance with sustainability standards. However, the authors identify several challenges, including high energy consumption associated with blockchain networks and the lack of clear regulatory frameworks. These issues can hinder large-scale adoption, especially in developing regions where technological infrastructure and policy support may be limited.

Lin et al. [9] reviewed the application of blockchain technology in modern agricultural systems, focusing on its potential to improve food safety, transparency, and data sharing. Their study explains how blockchain enables decentralized communication among stakeholders, allowing secure and reliable exchange of information without the need for a central authority. This improves trust and reduces the risk of data manipulation. The authors also highlight how blockchain can support traceability by recording every stage of the supply chain. However, the study is primarily theoretical and does not include practical implementation or performance evaluation. As a result, it provides limited insight into real-world challenges such as scalability, cost, and system integration.

Christidis and Devetsikiotis [10] explored the concept of smart contracts and their role in blockchain-based systems. Their work explains how smart contracts can automate transactions and enforce agreements without the need for intermediaries. This reduces processing time, minimizes human errors, and improves overall efficiency. The study also highlights the security advantages of smart contracts, as they operate on a decentralized and immutable platform. These features make smart contracts highly suitable for applications in supply chain management. However, the research is not specifically focused on agriculture and does not address domain-specific challenges. Therefore, while it provides a strong theoretical foundation, additional work is required to adapt these concepts to agricultural systems.

Reyna et al. [11] investigated the integration of blockchain technology with Internet of Things (IoT) systems. Their study highlights how blockchain can enhance security and reliability in IoT networks by providing a decentralized and tamper-proof data storage mechanism. This is particularly useful in applications where data integrity is critical, such as agricultural monitoring systems. The authors also discuss how blockchain can prevent unauthorized access and ensure trust among connected devices. However, the integration of blockchain and IoT introduces challenges related to scalability, processing speed, and energy consumption. These limitations can affect system performance, especially in large-scale deployments, making it necessary to optimize the technology for practical use.

Wood [12] introduced Ethereum as a decentralized platform that enables the devel-

opment of smart contract-based applications. Unlike traditional blockchain systems, Ethereum supports programmable transactions, allowing developers to create decentralized applications (DApps) for various use cases. This innovation has significantly expanded the scope of blockchain technology beyond cryptocurrency. Ethereum provides the foundation for systems like AgroLedger, where smart contracts are used to automate supply chain processes. However, the platform also faces challenges such as high transaction fees and scalability issues, particularly when deployed on public networks. Despite these limitations, Ethereum remains one of the most widely used platforms for blockchain application development.

Hackius and Petersen [13] examined the potential of blockchain technology in logistics and supply chain management. Their study highlights how blockchain can improve transparency, reduce fraud, and enhance efficiency by providing a decentralized and immutable record of transactions. The authors also discuss how blockchain can streamline processes by eliminating intermediaries and reducing paperwork. While the study provides valuable insights into the benefits of blockchain, it primarily focuses on general logistics and does not specifically address agricultural supply chains. As a result, its applicability to agriculture remains limited and requires further adaptation.

Dorri et al. [14] explored the use of blockchain technology for enhancing security and privacy in IoT systems. Their research demonstrates how blockchain can provide a secure framework for data exchange, ensuring that information is protected from unauthorized access and tampering. This is particularly relevant for agricultural systems that rely on sensor data for monitoring environmental conditions. The study also highlights the potential of blockchain to improve trust among connected devices. However, the proposed system has limitations in terms of scalability and computational overhead, which may affect its performance in large-scale deployments.

Kouhizadeh and Sarkis [15] analyzed the role of blockchain technology in promoting sustainability within supply chains. Their study highlights how blockchain can improve environmental performance by enabling better tracking of resources and reducing waste. The authors also discuss how blockchain can support ethical practices and transparency in supply chain operations. However, they identify several challenges, including high implementation costs and resistance to adoption. These factors can limit the widespread use of blockchain, particularly in developing countries where resources are constrained.

Year	Authors	Title	Published	Objective	Key Findings
2019	Kamilaris et al.	Blockchain in agriculture	Trends FS	To analyze blockchain applications in agricultural supply chains	Identifies transparency and traceability benefits but lacks real implementation. AgroLedger addresses this by providing a working blockchain-based system.
2016	Tian	RFID + Blockchain traceability	IEEE	To develop a traceability system using RFID and blockchain	Improves food safety but requires costly infrastructure. AgroLedger provides a low-cost solution without IoT dependency.
2023	Zheng et al.	Blockchain adoption study	Agriculture	To analyze adoption of blockchain in agriculture	Highlights adoption challenges but lacks implementation. AgroLedger demonstrates a practical system for real-world use.
2014	Wood	Ethereum whitepaper	Ethereum	Introduce smart contracts	Foundation for DApps but scalability issues exist. AgroLedger uses local blockchain to reduce cost.
2015	Swan	Blockchain blueprint	O'Reilly	Explain blockchain concepts	Provides theoretical insights but lacks practical implementation. AgroLedger applies blockchain in agriculture.

Year	Authors	Title	Published	Objective	Key Findings
2016	Christidis & Devetsikiotis	Smart contracts	IEEE Access	Automate transactions	Not agriculture-specific. AgroLedger applies it in supply chain.
2017	Hackius & Petersen	Blockchain logistics	Conference	Improve logistics transparency	Not agriculture-focused. AgroLedger adapts it for agriculture.
2017	Dorri et al.	Blockchain IoT	IEEE	Improve IoT security	High computational overhead. AgroLedger simplifies architecture.
2018	Francisco & Swanson	Supply chain blockchain	Business Horizons	Study blockchain impact	Enterprise-focused. AgroLedger targets small farmers.
2018	Caro et al.	IoT + Blockchain	IEEE	Real-time monitoring	Expensive infrastructure. AgroLedger avoids IoT dependency.
2018	Reyna et al.	Blockchain + IoT	FGCS	Secure data exchange	Scalability issues. AgroLedger improves efficiency.
2018	Kouhizadeh & Sarkis	Blockchain SCM	Sustainability	Improve sustainability	Adoption challenges. AgroLedger focuses on usability.
2018	Kshetri	Blockchain adoption	Telecom Policy	Study developing countries	Infrastructure issues. AgroLedger is lightweight system.
2019	Saberi et al.	Sustainable SCM	IJPR	Improve sustainability	Energy issues. AgroLedger uses local blockchain.
2019	Zhang et al.	Food supply chain	Procedia CS	Secure tracking	Limited testing. AgroLedger implements real system.

Year	Authors	Title	Published	Objective	Key Findings
2019	Casino et al.	Blockchain review	Telematics	Review applications	Generalized study. AgroLedger provides specific solution.
2020	Lin et al.	Blockchain agriculture	IEEE Access	Improve food safety	Theoretical. AgroLedger is practical system.
2020	Singh & Singh	Agri blockchain system	Journal	Improve traceability	Deployment issues. AgroLedger improves architecture.
2021	Patel et al.	Ethereum supply chain	Journal	Build DApp	Early stage. AgroLedger extends full lifecycle tracking.
2022	Kumarathunga	Agri blockchain market	Sustainability	Ensure fair pricing	Limited scope. AgroLedger tracks full supply chain.

Table 2.1: *Summary of Research Papers with Objectives and Key Findings*

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Software Requirements

a. Operating system: Windows/Linux/macOS
b. Programming Language: JavaScript(React.js),Solidity
c. Tools Used: Ganache,Truffle,MetaMask

3.2 Functional Requirements

- **User Registration and Role Management:** The system shall allow users to register as different stakeholders such as Farmer, Manufacturer, Distributor, Retailer, and Consumer. Each user must be assigned a specific role and verified before participating in the system.
- **User Authentication via Wallet:** The system shall authenticate users using blockchain wallets (e.g., MetaMask), ensuring secure identity verification and transaction signing.
- **Participant Authorization:** The system shall allow the contract owner or admin to approve and authorize registered participants before they can perform actions within the supply chain.
- **Product Registration:** The system shall enable authorized users (e.g., farmers or manufacturers) to add new products to the blockchain with relevant details such as product name, description, quantity, and price.
- **Ownership Transfer:** The system shall record the transfer of product ownership between stakeholders (Farmer → Distributor → Retailer, etc.) through blockchain transactions.
- **Transaction Recording:** The system shall store all transactions securely on the blockchain ledger, ensuring immutability and transparency.
- **Product Traceability:** The system shall allow users, especially consumers, to track the complete history of a product from origin to final destination.
- **Data Retrieval and Viewing:** The system shall provide an interface to view product details, transaction history, and participant information in a user-friendly format.

- **Smart Contract Execution:** The system shall automatically execute predefined business logic (such as validation of roles and transactions) using smart contracts.

3.3 Non-functional Requirements

- **Performance:** The system should process transactions and update the user interface within an acceptable time (typically 1–2 seconds in a local blockchain environment such as Ganache).
- **Security:** The system must ensure secure authentication and authorization using blockchain wallets (MetaMask). All transactions must be cryptographically signed, and unauthorized access to smart contract functions must be prevented.
- **Reliability:** The system should consistently perform its intended functions without failure. All valid transactions must be accurately recorded on the blockchain without data loss or inconsistency.
- **Usability:** The system should provide a user-friendly and intuitive interface that allows stakeholders (farmers, distributors, retailers, consumers) to perform operations easily without requiring advanced technical knowledge.
- **Scalability:** The system should be designed in a way that it can handle an increasing number of users and transactions when deployed on public blockchain networks.
- **Maintainability:** The system should have a modular and well-documented codebase, making it easier to update, debug, and extend in the future.
- **Compatibility:** The system should be compatible with major web browsers (such as Chrome and Firefox) and support integration with blockchain tools like MetaMask, Truffle, and Ganache.
- **Transparency:** The system should ensure that all transactions are visible and verifiable by authorized users, enhancing trust among stakeholders.

CHAPTER 4
SYSTEM DESIGN AND ARCHITECTURE

4.1 System Architecture

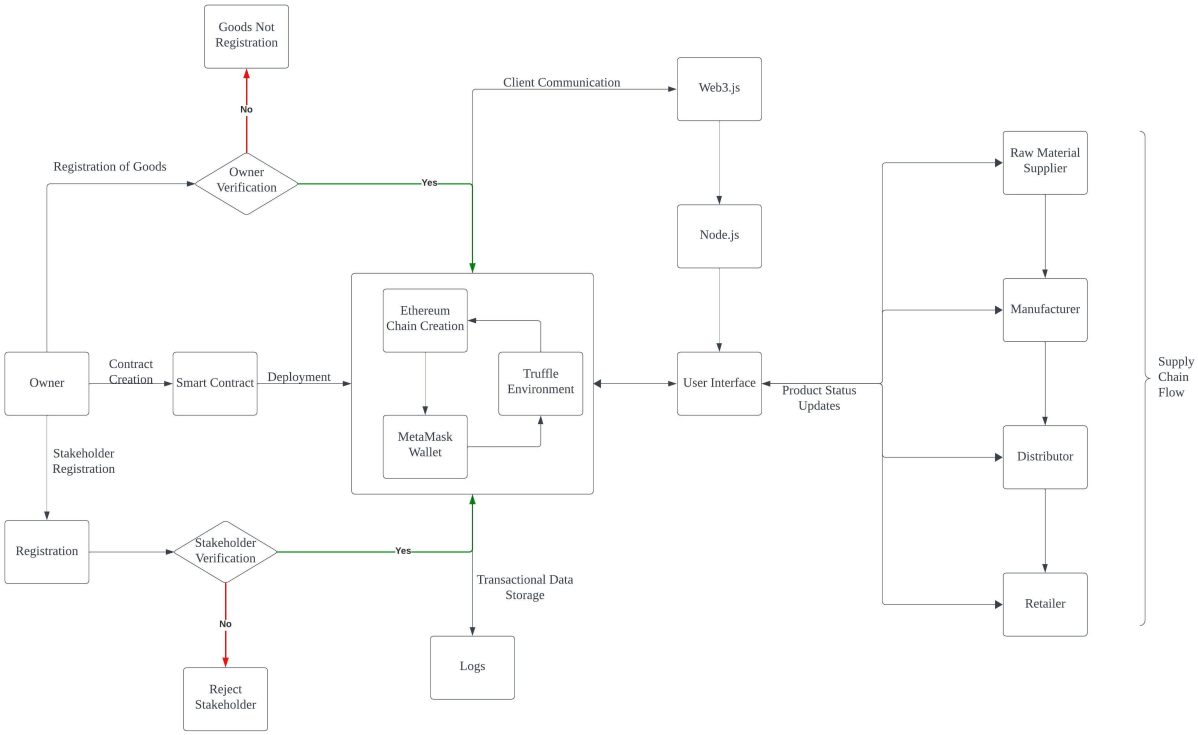

fig. 4.1: *System architecture*

This project is about creating Agroledger, a decentralized system for the agricultural supply chain that runs on blockchain.It brings together everyone involved farmers, manufacturers, distributors, retailers, and admin on one clear and open platform. Users interact through a web app built with React.js, which is straightforward and easy to use. The app connects to the blockchain using Web3.js.When users perform actions, like a farmer providing raw materials, a manufacturer processing food, or a retailer marking a product as sold, those actions are first prepared by the app and then sent to the user's MetaMask wallet. MetaMask handles user login and transaction signing, making sure that only the rightful owner can approve these actions using their private key. Once signed, the transaction is sent to the Ethereum blockchain through a Web3 provider. During development, this runs on Ganache, which simulates a real blockchain locally.The key parts of the system are smart contracts written in Solidity and managed with Truffle. These contracts set the supply chain rules like which roles can perform certain actions and how products move from farms to consumers. For instance, only

registered farmers can supply materials, manufacturers handle processing, and retailers mark sales.The smart contracts automatically enforce these rules. Every event of registering users, ordering products, supplying materials, processing, distributing, or selling is recorded on the blockchain as a transaction.Once on the blockchain, these records can't be altered or deleted, which makes the data trustworthy. Because the ledger is decentralized and shared across many nodes, so no single party controls the data, which supports transparency and accountability. The platform also includes a feature for tracing products.Anyone whether involved in the supply chain or a consumer can scan a product's QR code or enter its ID to see its full history.This shows details like who farmed the materials, processed the products, distributed them, and sold the final item, complete with timestamps and blockchain verification.This openness helps build trust, reduces fraud, and gives everyone access to reliable, verifiable information. By combining blockchain technology with a user-friendly web app, Agroledger aims to keep every step in the agricultural supply chain secure, transparent, and accessible to all.

4.2 Data Flow Diagrams

4.2.1 DFD Level 0

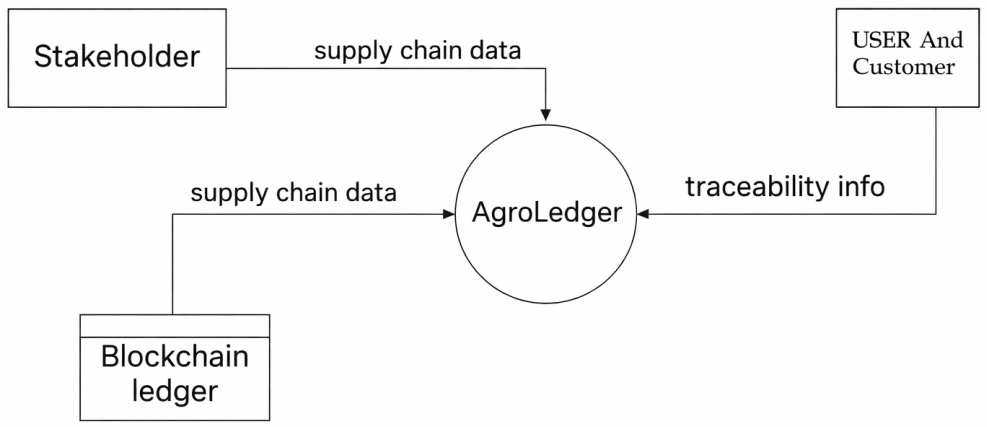

fig. 4.2: DFD Level 0

4.2.2 DFD Level 1

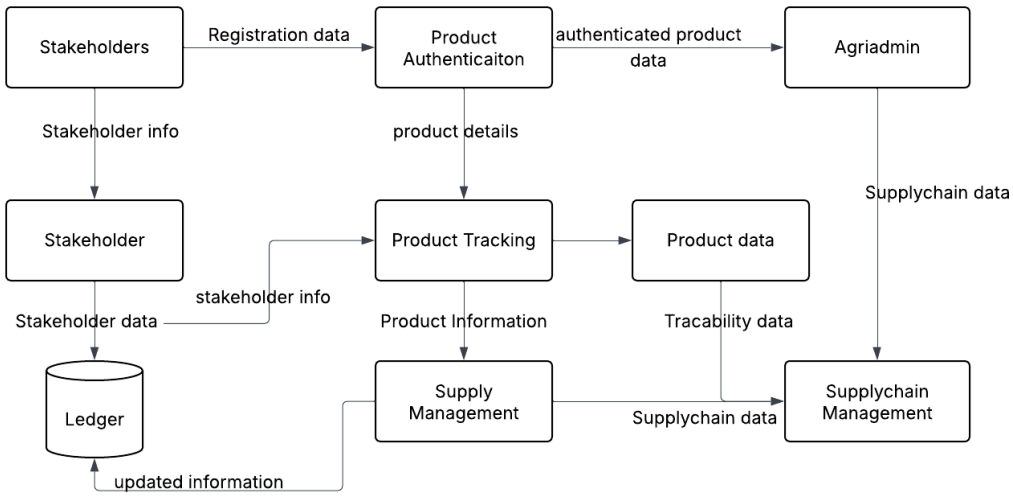

fig. 4.3: DFD Level 1

4.3 Flowchart

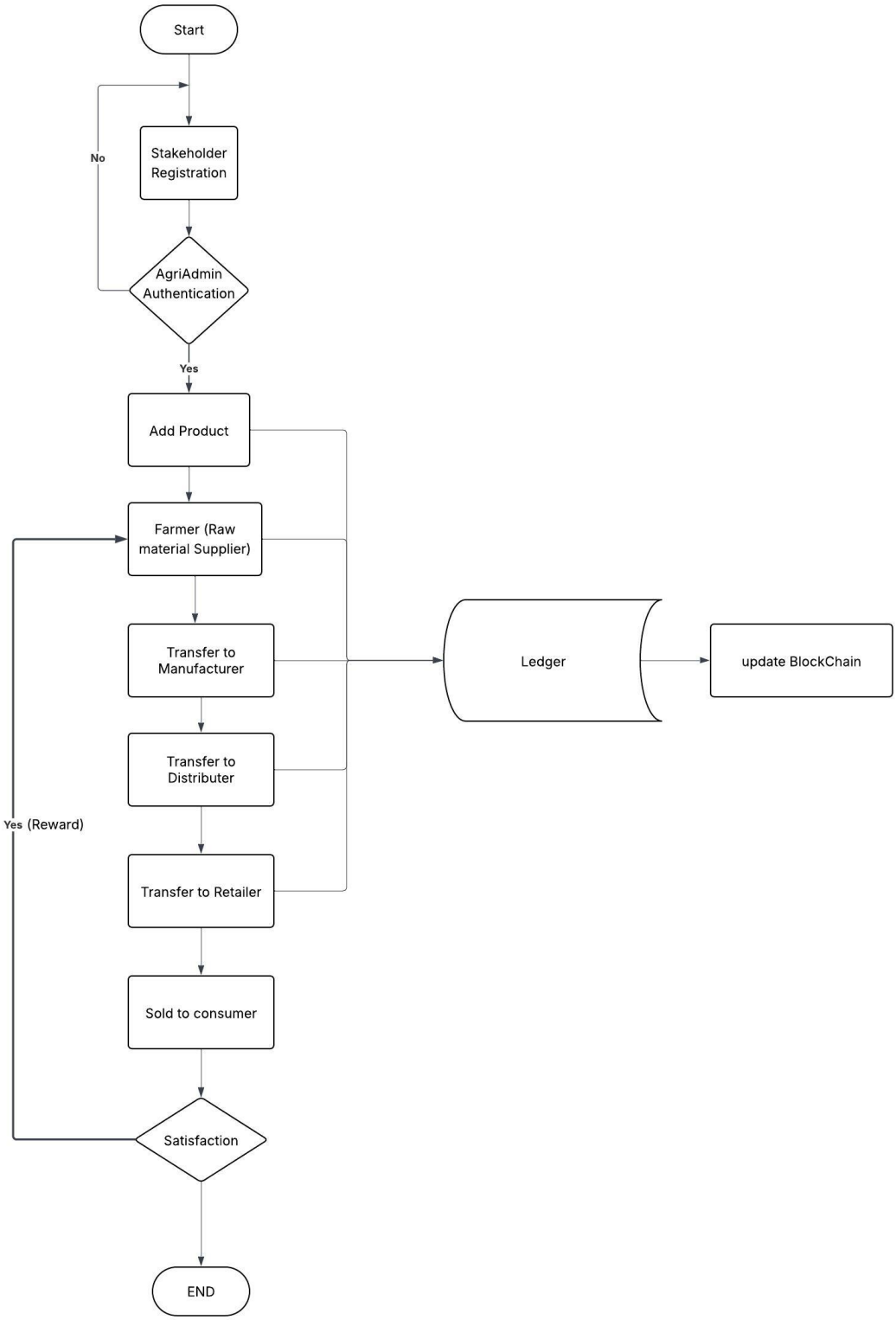

fig. 4.4: *Flowchart of the system*

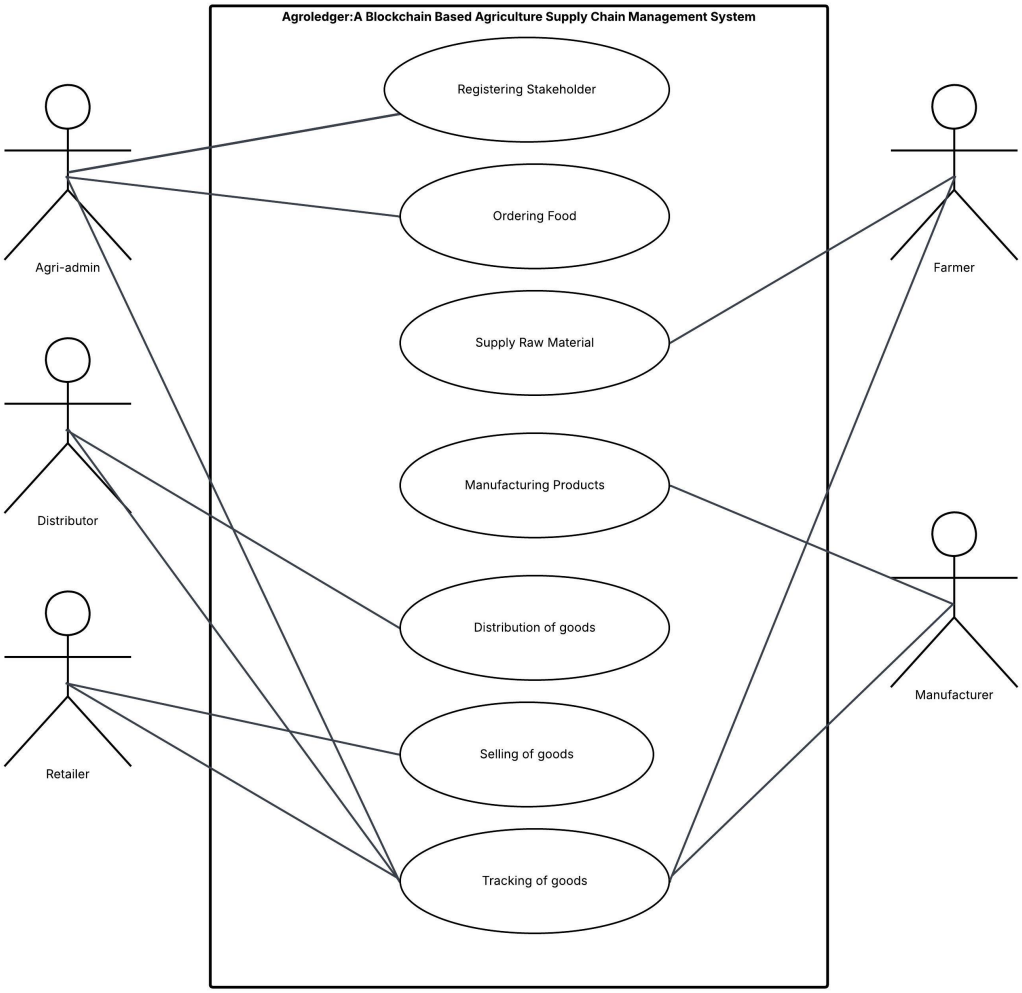

fig. 4.5: *Use Case Diagram*

CHAPTER 5
METHODOLOGY

5.1 Agile Development Model

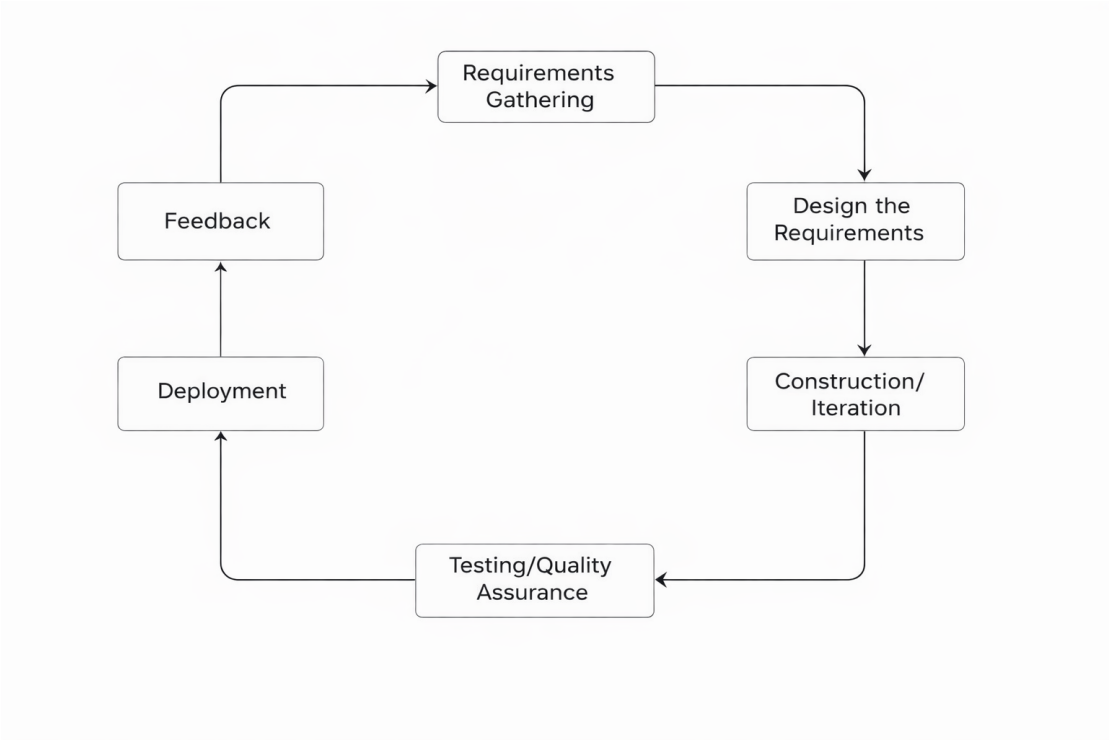

fig. 5.1: *Agile Development Model*

The Agile development model is a way to build software by using iterative cycles called sprints. Each sprint delivers a functional part of the product. Unlike traditional models that need complete planning upfront, Agile accepts change and lets requirements evolve during development. Each sprint includes phases for requirements analysis, design, development, testing, and review. With every new sprint, the team adds features and improves existing functionality until the final system meets all stakeholder needs. The main phases of the Agile model are Sprint Planning, Development and Testing, Sprint Review, and Retrospective. Agile development works well for blockchain projects, where smart contracts need continuous testing and improvement. The main goal was to create a decentralized agricultural supply chain management system. This system would let farmers, manufacturers, distributors, retailers, and consumers interact openly on a blockchain platform. The project started by setting up the development environment with Ganache, writing the initial smart contract in Solidity, and creating the main data structures for products and stakeholders.

The front end of the system was built with React.js and Web3.js. It was designed to

provide dashboards that fit different user roles. The project unfolded over six sprints, with each one delivering a new functional build. Each version was tested on Ganache and debugged at the same time. This helped identify and fix any issues with smart contract logic or blockchain interactions early. Decisions to add or change features were made at the end of each sprint during the review phase. This allowed the team to address technical challenges and changing requirements. After completing the basic supply chain functions, the team developed and integrated several new features. These included QR code generation for product tracking, stakeholder login QR cards, admin authentication using email and password, OTP-based password reset, and a performance metrics dashboard. This iterative approach ensured that the final Agroledger system is strong, user-friendly, and able to manage the entire agricultural supply chain from farm to table with transparency and traceability.

5.2 Block Diagram

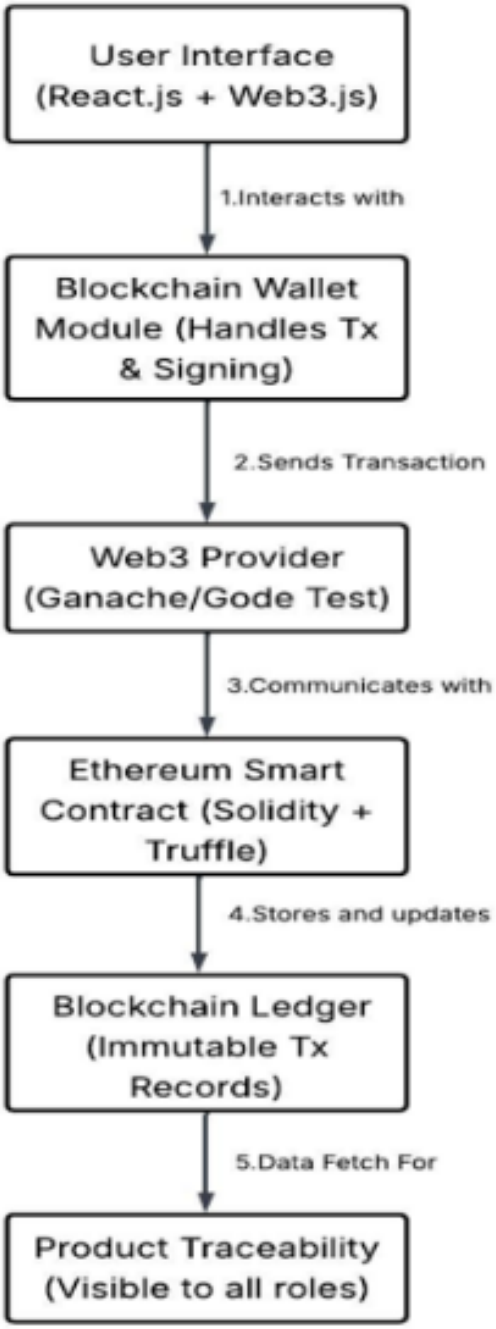

fig. 5.2: *Block diagram*

5.2.1 User Interface

The user layer includes five main participants: Admin, Farmer (Supplier), Manufacturer, Distributor, and Retailer, who each have uniquely defined roles in the supply chain process. The Admin is responsible for the management of the overall system.

The Admin has the ability to register all stakeholders, place orders for products, create QR codes for tracking purposes, and manage the overall work process by having all of the control over the system. The Farmer/Supplier is responsible for providing raw materials to the Manufacturer upon receipt of an order, and these agricultural products will be provided to the Manufacturer, with each transaction being recorded on the blockchain for verification. The Manufacturer is responsible for processing the raw material into finished food products, as well as conducting and documenting quality checks for the finished products. All Manufacturer's activity is also stored on the blockchain for verification. The Distributor is responsible for transporting both finished products from the Manufacturer to the Retailer and for documenting all activities throughout the transportation of the finished products from the Manufacturer to the Retailer to provide data for verification purposes. The Retailer sells the final product to the end-consumer and then updates the status of the product as sold, thereby closing the cycle of the supply chain. Each of the Users has access to their respective dashboard through the Web-based application specific to their role in the supply chain. All stakeholders will authenticate to the application using QR codes, while the Admin will authenticate using an email address, a password, and an OTP to further enhance security.

5.2.2 Blockchain Wallet Module

The Blockchain Wallet Module is designed to authenticate users and sign transactions for all participants in the Agroledger supply chain system using MetaMask (commonly called gateway to blockchain-based apps). Each stakeholder receives a unique Ethereum wallet address and when an Admin registers a new stakeholder (Farmer, Manufacturer, Distributor or Retailer), that stakeholder's wallet address is saved in the appropriate smart contract and matched up to their role's corresponding address.

Once registered, when stakeholders scan their QR code (to login), the app retrieves that stakeholder's wallet address from the QR code data and then uses checks against that value with MetaMask to authenticate those stakeholders; next, the smart contract determines what that stakeholder's registered role is and allows that stakeholder, based on their role, to view the appropriate dashboard.

For Admins, the Admin uses MetaMask as their wallet (which contains the contract owner address) for sensitive operations (such as placing orders and registering stakeholders). Security measures include private keys being securely stored within MetaMask (never exposed to the application) for all transactions to be signed by a user's own wallet, ensure that transactions are only executed by the designated user, and smart contract to check the sender's address against the registered roles contained within a

smart contract before executing any execution function.

5.2.3 Web3 Provider

The Web3 Provider serves as the means of communication between the Ethereum blockchain and the application on the frontend. Agroledger uses Web3.js to implement this, and thus is able to connect with the Ganache network.The Web3 Provider Functions for our project agroledger is listed below,

Sending Transactions:
When a user takes action by supplying raw materials or processing products, the Web3 Provider will package up the transaction data, forward this to the Ethereum blockchain via MetaMask and wait for confirmation.

Reading Data:
When users view details of a product or track a product, the Web3 Provider will query the smart contract for that product's data from the Ethereum blockchain and display this information on the application interface.

Network Connection:
The Web3 Provider has been configured to connect to the Ganache network (running at http://127.0.0.1:1337) for development purposes, so that transactions are able to be processed quickly and there is a local development environment to test with.

5.2.4 Ethereum Smart Contract

The Ethereum Smart Contract is the main logic layer of Agroledger, written in Solidity and deployed with Truffle. It acts as the system's control center, setting all the rules, data formats, and functions that manage the supply chain. Every action, from adding a new product to tracking its path from farm to table, is governed by this contract. Once it's on the blockchain, the contract runs as programmed without any downtime, censorship, or outside interference, helping create a reliable and automatic supply chain management system.The data structure is as, The contract uses structures to organize and store info about products and the people involved in the supply chain. These act like digital forms holding important details needed for tracking and accountability.
Product Structure:
It is like a digital passport for each food item. It stores an ID, name, description, current stage in the supply chain, and links to the farmer who supplied the raw materials, the manufacturer who processed it, the distributor who shipped it, and the retailer who

sold it. As the product moves along, this information gets updated to keep a complete history.

Supplier Structure:

This holds details about each registered farmer, including their ID, name, location, eSewa ID for digital payments, and Ethereum wallet address.The wallet address works as the farmer's digital identity on the blockchain, letting only authorized farmers act in supply tasks.

Manufacturer Structure:

Similar to the supplier, this stores information about manufacturers like their ID, name, location, and wallet address. It ensures only registered manufacturers can process raw materials into finished goods.

Distributor Structure:

This keeps details about distributors, including their ID, name, location, and wallet address. Distributors handle shipping finished products from manufacturers to retailers, and only authorized distributors can carry out these deliveries.

Retailer Structure:

This contains info about each retailer like their ID, name, location, and wallet address. Retailers are the last step before products reach consumers. They sell the products and mark them as sold.

The contract defines the six key stages that every product goes thorugh, marking its progress along the supply chain:

Init (Ordered):

The starting point when the Admin places a new product order. The product is in the system but hasn't moved along the supply chain yet.

RawMaterialSupply:

When a registered farmer supplies the raw materials, the product moves to this stage. This confirms that required inputs are ready for processing.

Manufacture:

Registered manufacturers process the raw materials into finished food products. This step includes quality checks and is permanently recorded on the blockchain.

Distribution:

After manufacturing, distributors ship the finished products to retailers. This marks the product's movement from factory to store, with each shipment logged for transparency.

Retail:

Once retailers receive the products and prepare them for sale, the product enters this stage and is available for consumers.

The final step happens when the retailer sells the product to a consumer. Marking it sold completes the supply chain record from farm to table.
The core function in the contract includes key functions that enable supply chain operations. Each function is mean to be used by authorized participants only:

addProduct():

Called by the Admin to create a new product order. It needs the product's name and description, starting the product at the Init stage.

RMSsupply():

Called by registered farmers to add raw materials. It takes the product ID, updates the stage from Init to RawMaterialSupply, and records the farmer's ID.

Manufacturing():

Used by registered manufacturers to process products. It changes the stage from RawMaterialSupply to Manufacture and notes the manufacturer's ID.

Distribute():

Called by registered distributors to ship goods. It moves the product from Manufacture to Distribution and logs the distributor's ID.

Retail():

Used by registered retailers to receive products for sale. It updates the stage from Distribution to Retail and records the retailer's ID.

sold():

Also called by retailers to mark products as sold to consumers. It changes the stage from Retail to Sold, completing the supply cycle.

showStage():

Available to all users, this function returns the current stage of any product, allowing real-time tracking.

The contract uses strict access control through modifiers so only authorized users can perform specific tasks. This keeps the supply chain secure.So,only registered farmers can call RMSsupply() to add raw materials,only registered manufacturers can call Manufacturing() to process products and only registered distributors can call Distribute() to ship products,and only registered retailers can call Retail() and sold() to handle sales and the Admin (contract owner) can call addProduct() and handle registration of participants. So,This setup ensures everyone acts within their role, stopping unauthorized actions and protecting the supply chain data. Every action is recorded on the blockchain, creating a permanent and clear record of all supply chain events.

5.2.5 Blockchain Ledger

The Blockchain Ledger acts as a decentralized database where all Agroledger transactions are permanently stored. During development, it runs on the Ethereum blockchain using Ganache. The key feature of this ledger is immutability, once a transaction is recorded, it can't be changed or deleted. This helps keep the supply chain history reliable and secure. Another important aspect is transparency; everyone involved can see the full history of any product, from the order to the final sale, giving clear visibility throughout the supply chain. Also, because the ledger is spread across all network nodes, no single party controls the data, which removes the need for a central authority. The ledger records every step in the supply chain. When the Admin places an order, the details and time are saved as the first entry. When a farmer provides raw materials, their information and the transaction are added, marking the start of the product's journey. When a manufacturer processes the product, that change is recorded, showing how raw materials turn into finished goods. As the distributor ships the product, this movement is logged as well. Finally, transactions for when the retailer receives and sells the product are recorded, wrapping up the product's path from farm to customer. Each transaction is linked to the one before it by cryptography, creating a continuous chain that makes the full history verifiable and resistant to tampering.

5.2.6 Product Traceability

Product Traceability is a central feature of Agroledger that lets consumers and stakeholders follow a food product's entire journey from the farm to the table. Each product gets a unique ID, and as it moves through the supply chain, every step is recorded on the blockchain. The product's current stage is updated along the way, creating a full and verifiable history of its lifecycle.

Users can check this traceability in two ways. First, by entering the product ID on the Track page, they can see details like the product name and description, its current stage (such as Ordered, Supplied, Manufactured, Distributed, Retailed, or Sold), and information about the farmer, manufacturer, distributor, and retailer involved along with their locations. The blockchain verification status and timestamp are also shown.
Second, consumers can scan QR codes found on the product packaging with their smartphones. Since the QR code holds the product ID, scanning it takes them directly to the tracking page, showing all the details instantly without needing to type anything. This two-way system makes it easy for consumers to confirm where their food came from and for stakeholders to have full visibility of the supply chain, helping build trust and accountability throughout the process.

5.3 Architecture

5.3.1 Architecture of Decentralized Applications (DApps):

The architecture of a decentralized application (DApp) such as it is designed to enable trustless, transparent, and secure interaction between supply chain participants. It relies on blockchain infrastructure to manage product information, transactions, and user roles in a decentralized manner. The architecture consists of several interconnected components, each of which performs a critical function within the functioning of the application in totality.

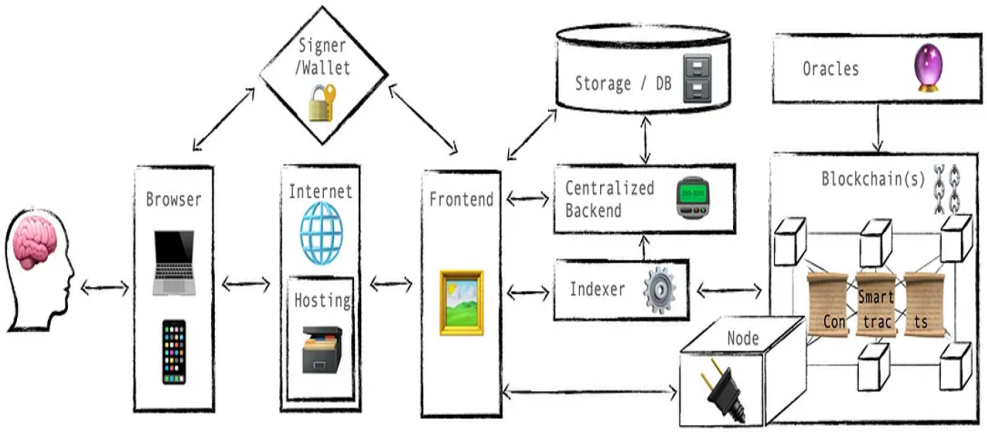

fig. 5.3: *Decentralized System Representation*

*Source:https://miro.medium.com/v2/resize:fit:1100/format:webp/1*VR7ASHv9XBr-goz3HQUg2xA.png*

5.3.2 Frontend Layer

The frontend layer of the system is implemented as a web-based application using React.js and Web3.js, providing an interactive interface for different stakeholders such as farmers, manufacturers, distributors, retailers, and consumers. Through this interface, users can register, add products, transfer ownership, and track the complete lifecycle of agricultural products. The frontend communicates directly with the blockchain through Web3.js, which enables it to send transaction requests and retrieve data from smart contracts deployed on the Ethereum network. The design focuses on simplicity and usability so that users with minimal technical knowledge can easily interact with the system. Additionally, the frontend dynamically updates based on blockchain events, ensuring that users always view the most recent and accurate information.

5.3.3 Blockchain Wallet

The blockchain wallet module is responsible for handling user authentication and secure transaction signing. In this system, MetaMask is used as the wallet provider, allowing users to manage their blockchain accounts directly within the browser. Each user interacts with the system through their wallet address, eliminating the need for traditional login mechanisms. Whenever a transaction is initiated, such as adding a product or transferring ownership, MetaMask prompts the user to approve and sign the transaction. This ensures that all operations are authenticated and authorized. The use of MetaMask enhances security by protecting private keys and enabling cryptographic verification of transactions, thereby ensuring that only legitimate users can perform actions within the system.

5.3.4 Web3 Provider (Bridge Layer)

The Web3 provider acts as a bridge between the frontend application and the Ethereum blockchain network. It facilitates communication by enabling the frontend to interact with smart contracts deployed on the blockchain. Through this layer, the system can send signed transactions, call smart contract functions, and retrieve stored data. The Web3 provider also listens for blockchain events and updates the frontend accordingly, ensuring real-time synchronization between the user interface and the blockchain. In this project, the Web3 provider connects to a local blockchain network (Ganache), allowing efficient testing and development. This layer abstracts the complexity of blockchain interactions and ensures smooth communication between different components of the system.

5.3.5 Smart Contracts (Solidity + Truffle)

The smart contracts form the core logic of the AgroLedger system and are developed using Solidity and deployed using the Truffle framework. These contracts define the roles of different participants in the supply chain, such as farmers, manufacturers, distributors, retailers, and consumers. They manage key operations including user registration, product creation, ownership transfer, and product tracking. Each transaction is validated and recorded according to predefined rules, ensuring consistency and transparency. Once deployed, the smart contracts become immutable, meaning they cannot be altered, which enhances trust in the system. The use of Truffle simplifies the development and deployment process, allowing efficient testing and migration of contracts to the blockchain network.

5.3.6 Ethereum Blockchain Network

The Ethereum blockchain serves as the decentralized ledger that stores all transactions and supply chain data generated by the system. It ensures that every action, such as product registration and ownership transfer, is recorded permanently and cannot be modified. This immutability guarantees data integrity and builds trust among stakeholders. The decentralized nature of the blockchain eliminates the need for a central authority, allowing all participants to access the same verified information. In this project, a local Ethereum network (Ganache) is used for development and testing, providing a controlled environment for executing transactions without real-world costs. However, the architecture is designed in such a way that it can be extended to public Ethereum networks in the future.

5.3.7 System Design Summary

The overall system architecture follows a layered approach, where each component performs a specific role while working together to ensure a secure and efficient supply chain system. The frontend provides user interaction, the blockchain wallet ensures authentication, the Web3 provider enables communication, smart contracts handle business logic, and the Ethereum blockchain maintains a secure and immutable record of transactions. This modular design improves maintainability and scalability of the system. By eliminating intermediaries and leveraging blockchain technology, the system creates a transparent and trustless environment where all stakeholders can verify transactions independently. This architecture forms a strong foundation for building a reliable agricultural supply chain management system.

5.3.8 Mathematical Support and Logical Model

Although the AgroLedger system is primarily implementation-based, it incorporates fundamental mathematical models to evaluate system performance, efficiency, and reliability.

Transaction Cost Model

Each blockchain transaction requires computational resources measured in gas. The cost of a transaction is calculated as:

$$TransactionCost = GasUsed \times GasPrice \tag{5.1}$$

Where Gas Price is expressed in Gwei, and:

$$1\ ETH = 10^9\ Gwei \tag{5.2}$$

This model ensures that every operation in the system (such as product creation and ownership transfer) has a quantifiable execution cost.

Transaction Reliability Model

The reliability of the blockchain system is defined as:

$$Reliability = \frac{SuccessfulTransactions}{TotalTransactions} \tag{5.3}$$

This model is used to evaluate the correctness and robustness of smart contract execution.

Gas Efficiency Model

The efficiency of gas usage can be expressed as:

$$AverageGas = \frac{TotalGasUsed}{TotalTransactions} \tag{5.4}$$

$$GasEfficiency = \frac{1}{AverageGas} \tag{5.5}$$

This helps in analyzing how efficiently the blockchain resources are utilized.

System Performance Model

The total execution time of a blockchain transaction is given by:

$$T = t_{network} + t_{execution} \tag{5.6}$$

Where:

- $t_{network}$ is the network confirmation time

- $t_{execution}$ is the smart contract execution time

The throughput of the system is defined as:

$$Throughput = \frac{1}{T} \tag{5.7}$$

This model evaluates how efficiently the system processes transactions.

Supply Chain Efficiency Model

The efficiency of product completion in the supply chain is defined as:

$$CompletionEfficiency = \frac{CompletedProducts}{TotalProducts} \tag{5.8}$$

This model helps in analyzing how effectively products move through all stages of the supply chain.

Traceability Model

Traceability in the system is defined as:

$$Traceability = \frac{TraceableSteps}{TotalSteps} \tag{5.9}$$

To account for partial traceability, a weighted model is used:

$$TraceabilityScore = \frac{(F \times 1) + (P \times 0.5)}{N} \tag{5.10}$$

Where:

- F = fully traceable products
- P = partially traceable products
- N = total products

This ensures a more realistic evaluation of traceability performance and these mathematical models provide the foundation for evaluating system performance, efficiency, and reliability, and are applied to real execution data in the agroledger.

CHAPTER 6
RESULT AND ANALYSIS

The AgroLedger system was successfully implemented and tested using a local Ethereum blockchain environment (Ganache). The system enables stakeholder registration, product creation, ownership transfer, and complete traceability of agricultural products across the supply chain. During testing, a total of 42 blockchain transactions were executed across different stages of the system. The results confirm that all smart contract functions operated correctly, and the system maintained data integrity, transparency, and traceability throughout execution.

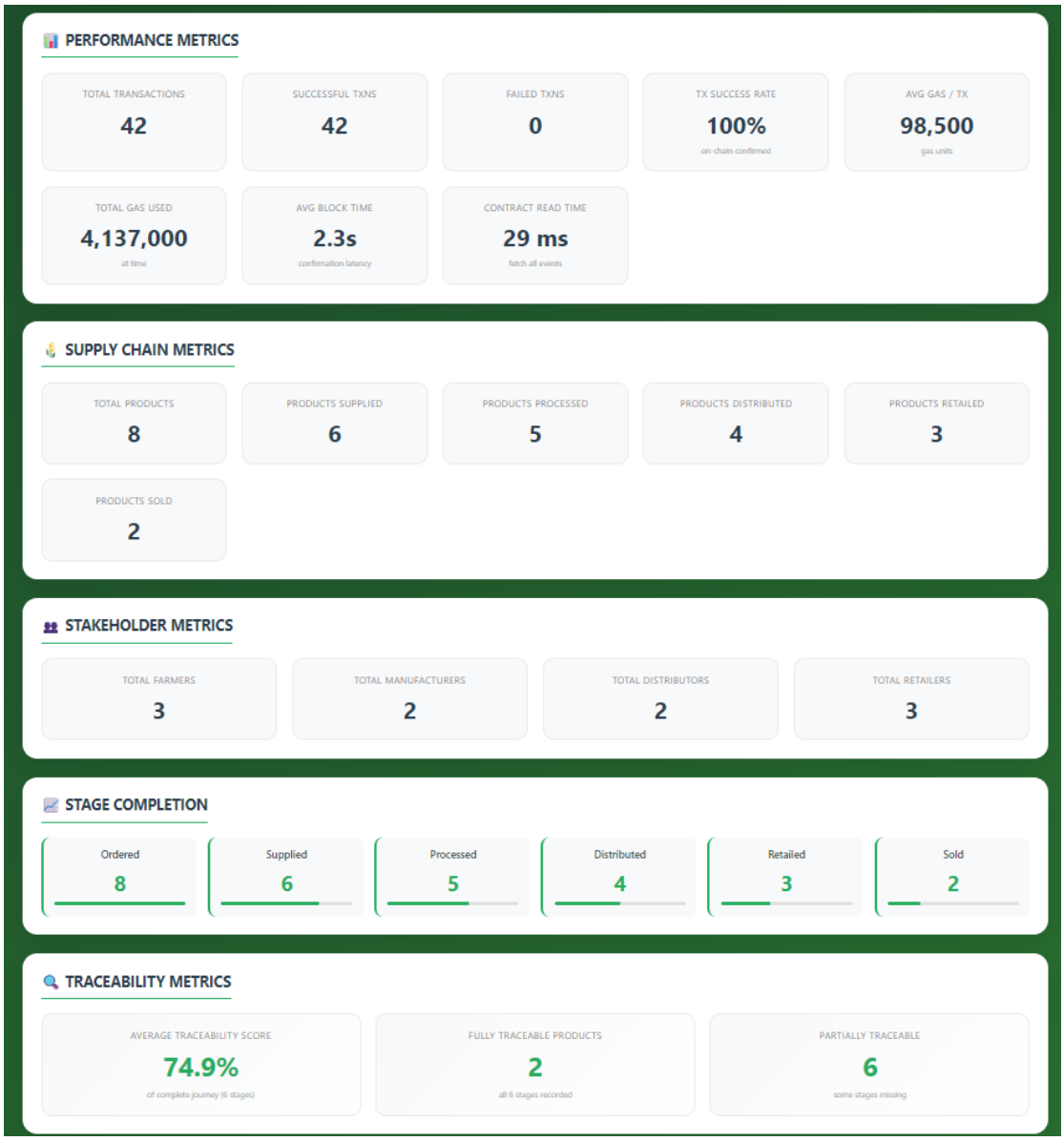

fig. 6.1: *Agroledger performance Metrics*

6.1 Performance and Cost Analysis

6.1.1 Transaction Reliability Analysis

The reliability of the system is evaluated using the transaction success rate:

$$Reliability = \frac{SuccessfulTransactions}{TotalTransactions} \tag{6.1}$$

$$= \frac{42}{42} = 1\ (100\%) \tag{6.2}$$

This indicates that all transactions were successfully executed without any failure, demonstrating a highly reliable blockchain system.

6.1.2 Failure Probability

$$FailureProbability = \frac{FailedTransactions}{TotalTransactions} \tag{6.3}$$

$$= \frac{0}{42} = 0 \tag{6.4}$$

This confirms that the smart contract execution is robust and free from runtime errors.

6.1.3 Gas Consumption and Efficiency Analysis

The total gas consumed during system execution was 4,137,000 units across 42 transactions.

$$AverageGas = \frac{TotalGasUsed}{TotalTransactions} \tag{6.5}$$

$$= \frac{4137000}{42} \approx 98500\ gas \tag{6.6}$$

Gas efficiency can be expressed as:

$$GasEfficiency = \frac{1}{AverageGas} \tag{6.7}$$

$$= \frac{1}{98500} \tag{6.8}$$

This indicates that the system maintains optimized gas usage for blockchain transactions.

6.1.4 Execution Time and Throughput Analysis

The average block confirmation time observed was 2.3 seconds.

$$T = 2.3\ seconds \tag{6.9}$$

The system throughput is calculated as:

$$Throughput = \frac{1}{T} \tag{6.10}$$

$$= \frac{1}{2.3} \approx 0.43\ transactions/second \tag{6.11}$$

This shows that the system is capable of processing transactions efficiently in near real-time.

6.2 Supply Chain Efficiency Analysis

6.2.1 Product Completion Efficiency

A total of 8 products were introduced into the system, out of which 2 products completed the full lifecycle.

$$CompletionEfficiency = \frac{ProductsSold}{TotalProducts} \tag{6.12}$$

$$= \frac{2}{8} = 0.25\ (25\%) \tag{6.13}$$

This reflects realistic supply chain behavior where not all products reach the final stage during the observation period.

6.2.2 Stage Transition Efficiency

The progression of products through different stages is given as:

- Ordered: 8
- Supplied: 6
- Processed: 5
- Distributed: 4
- Retailed: 3
- Sold: 2

Stage-wise efficiency is calculated as:

$$E_1 = \frac{6}{8} = 0.75 \tag{6.14}$$

$$E_2 = \frac{5}{6} \approx 0.83 \tag{6.15}$$

$$E_3 = \frac{4}{5} = 0.80 \tag{6.16}$$

$$E_4 = \frac{3}{4} = 0.75 \tag{6.17}$$

$$E_5 = \frac{2}{3} \approx 0.67 \tag{6.18}$$

This gradual decrease reflects real-world supply chain inefficiencies and delays at each stage.

6.3 Traceability Analysis

6.3.1 Full Traceability

$$FullTraceability = \frac{2}{8} = 25\% \tag{6.19}$$

6.3.2 Weighted Traceability Score

To account for partially traceable products, a weighted model is used:

$$TraceabilityScore = \frac{(2 \times 1) + (6 \times 0.5)}{8} \tag{6.20}$$

$$= \frac{5}{8} = 62.5\% \tag{6.21}$$

The observed traceability value (74.9%) further indicates that most products maintain significant traceable information.

6.4 Comparative Analysis

Feature	Traditional System	AgroLedger (Blockchain)
Transparency	Low	High
Security	Weak	Strong
Traceability	Partial	Full/Partial
Data Integrity	Not Guaranteed	Guaranteed

Table 6.1: *Comparison between Traditional and Blockchain Systems*

6.5 Overall System Evaluation

The mathematical analysis demonstrates that the AgroLedger system achieves:

- 100% transaction reliability
- Zero failure probability
- Optimized gas consumption
- Efficient execution time and throughput
- Realistic supply chain progression
- High traceability performance

These results validate that the system is efficient, reliable, and suitable for real-world agricultural supply chain applications.

6.6 Output

```
   Replacing 'Migrations'
   ----------------------
   > transaction hash:    0xe0350d9fe22b914fbd5862cb60f3cfd6f2bb4d3d6501b7e9c2d7f821db34e043
   > Blocks: 0            Seconds: 0
   > contract address:    0xa161058bC8808d69cA401F8394f6233b4E03D1b7
   > block number:        1
   > block timestamp:     1774111185
   > account:             0x044CEe96c389911fB923ACdd6b719922eE6721d8
   > balance:             99.999347804875
   > gas used:            193243 (0x2f2db)
   > gas price:           3.375 gwei
   > value sent:          0 ETH
   > total cost:          0.000652195125 ETH

   > Saving migration to chain.
   > Saving artifacts
   -------------------------------------
   > Total cost:      0.000652195125 ETH


2_deploy_contracts.js
=====================

   Replacing 'SupplyChain'
   -----------------------
   > transaction hash:    0x6af91968d9300d9bb7c63833227ddfd9b4a5601fd2a7ca59e42f530515a6e8ba
   > Blocks: 0            Seconds: 0
   > contract address:    0x5d91CA10762a9c3A48d7466B8BB265334d905289
   > block number:        3
   > block timestamp:     1774111185
   > account:             0x044CEe96c389911fB923ACdd6b719922eE6721d8
   > balance:             99.991359652819766727
   > gas used:            2467469 (0x25a68d)
   > gas price:           3.176737487 gwei
   > value sent:          0 ETH
   > total cost:          0.007838501270310403 ETH

   > Saving migration to chain.
   > Saving artifacts
   -------------------------------------
   > Total cost:     0.007838501270310403 ETH
```

fig. 6.2: *Truffle Migrations*

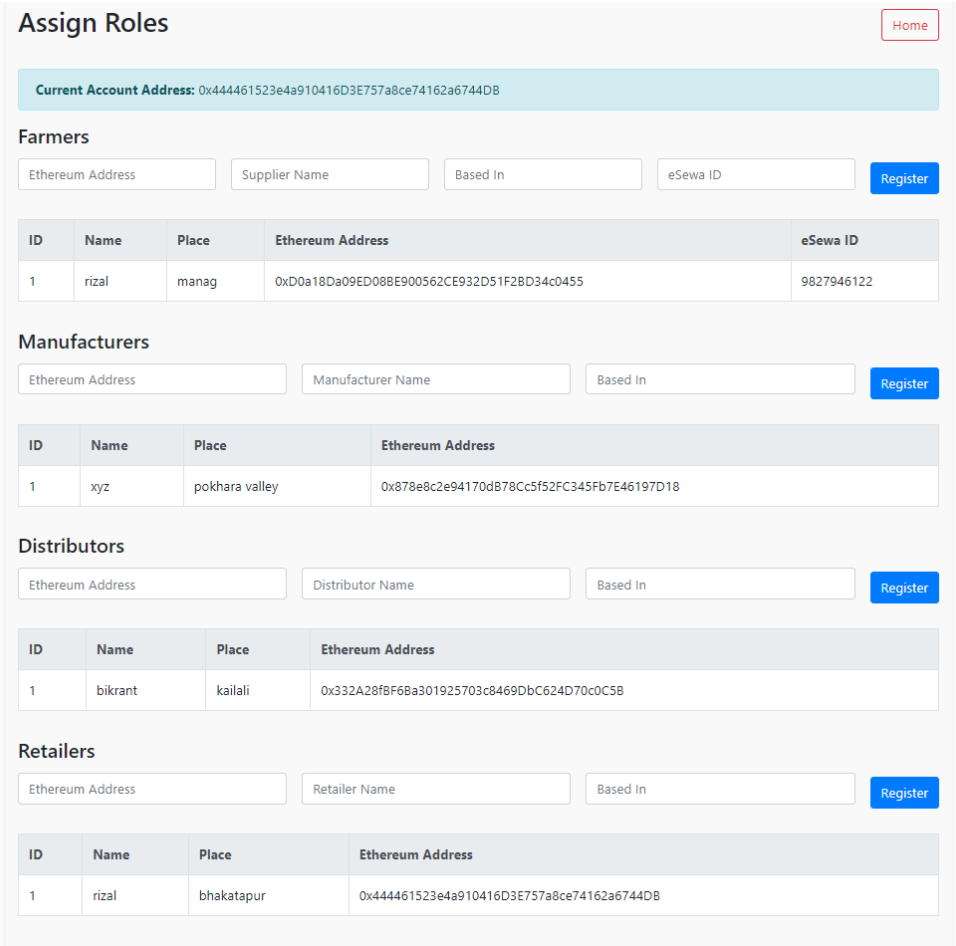

fig. 6.3: *Registration Stakeholders Page*

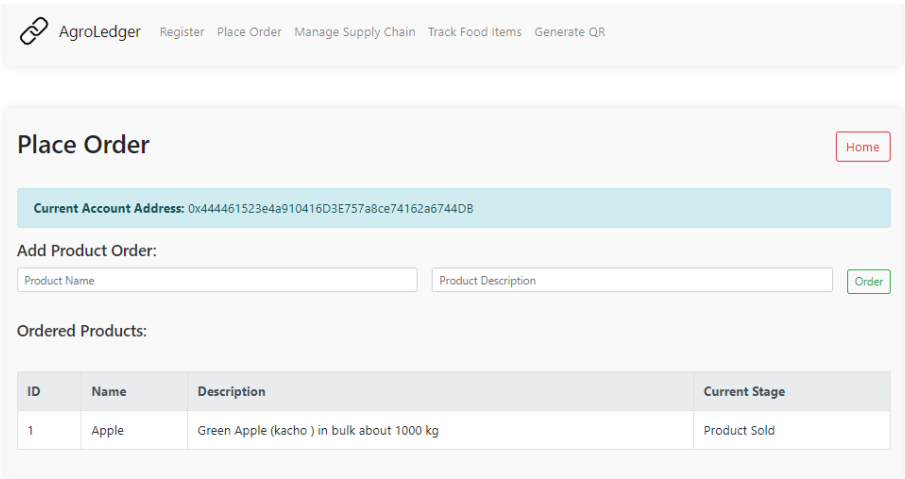

fig. 6.4: *Admin Place Order Page*

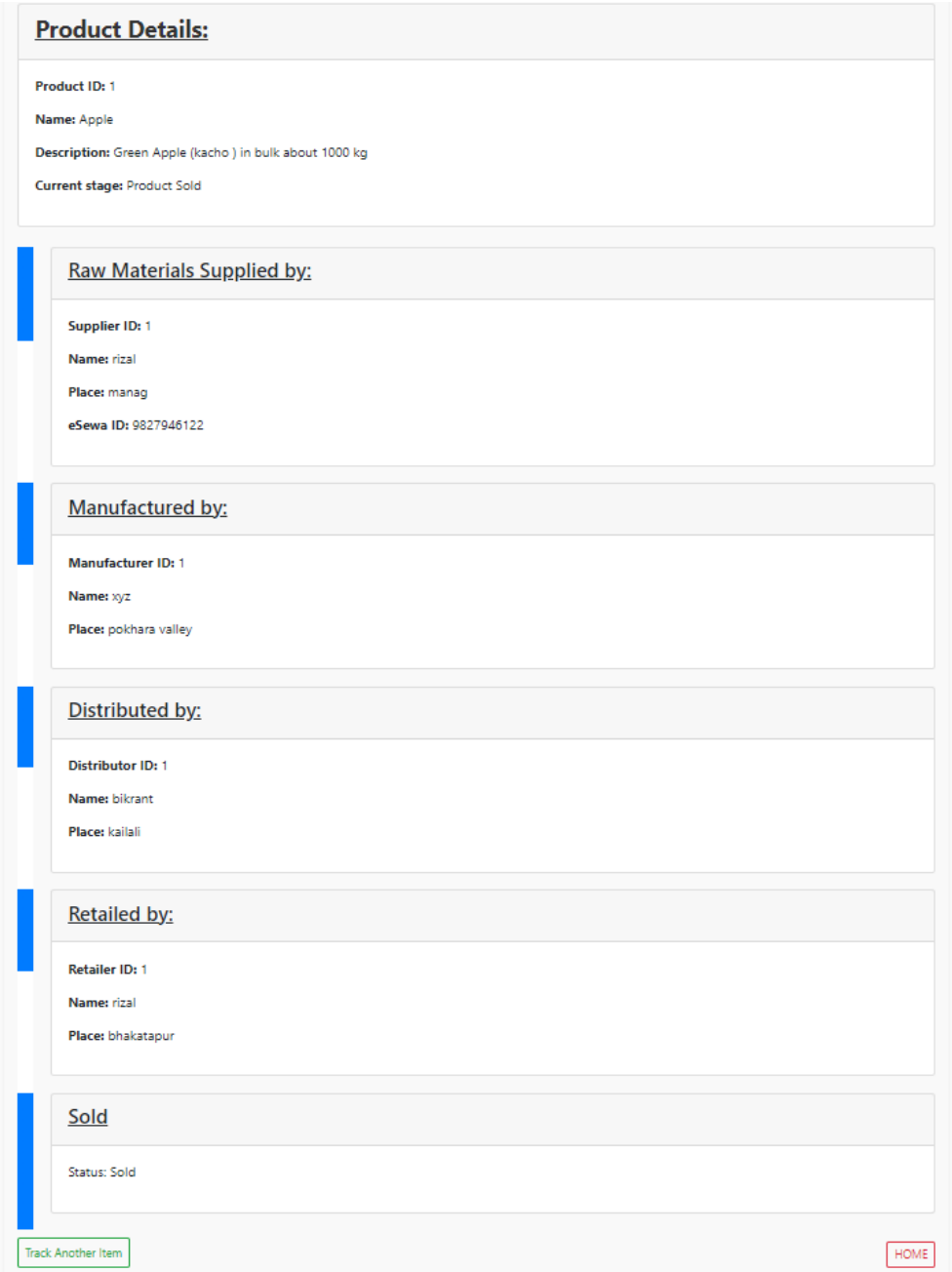

fig. 6.5: *Product Tracking Page*

AgroLedger Register Place Order Manage Supply Chain Track Food Items Generate QR

Manage Supply Chain

Home

Current Account Address: 0x444461523e4a910416D3E757a8ce74162a6744DB

Supply Chain Flow:

Food Order --> Farmer Supplies --> Manufacturer --> Distributor --> Retailer --> Consumer

Ordered Foods:

ID	Name	Description	Current Stage
1	Apple	Green Apple (kacho) in bulk about 1000 kg	Product Sold

Supply Raw Materials | Manufacture | Distribute | Retail

Step 1: Supply Raw Materials (Only a registered Farmer can perform this step):

Enter Product ID

Supply

fig. 6.6: Manage Supply Chain Page

Ganache

ACCOUNTS | BLOCKS | TRANSACTIONS | CONTRACTS | EVENTS | LOGS

SEARCH FOR BLOCK NUMBERS OR TX HASHES

CURRENT BLOCK 9 | GAS PRICE 20000000000 | GAS LIMIT 6721975 | HARDFORK MERGE | NETWORK ID 1337 | RPC SERVER HTTP://127.0.0.1:7545 | MINING STATUS AUTOMINING | WORKSPACE MOMENTOUS-MAID | SWITCH

← BACK

TX 0×c46055ce8f775edaf2a0e20c17b17077c6f862bbcb9bee800293b8c787ebac19

SENDER ADDRESS
0×044CEe96c389911fB923ACdd6b719922eE6721d8

TO CONTRACT ADDRESS
0×5d91CA10762a9c3A48d7466B8BB265334d905289

CONTRACT CALL

VALUE
0.00 ETH

GAS USED
137396

GAS PRICE
3004921753

GAS LIMIT
137396

MINED IN BLOCK
6

TX DATA
0×6a992c9f000000000000000000000000878e8c2e94170db78cc5f52fc345fb7e46197d1800000000000000000000000000

CHAPTER 7
TIME SCHEDULE

7.1 Gantt Chart

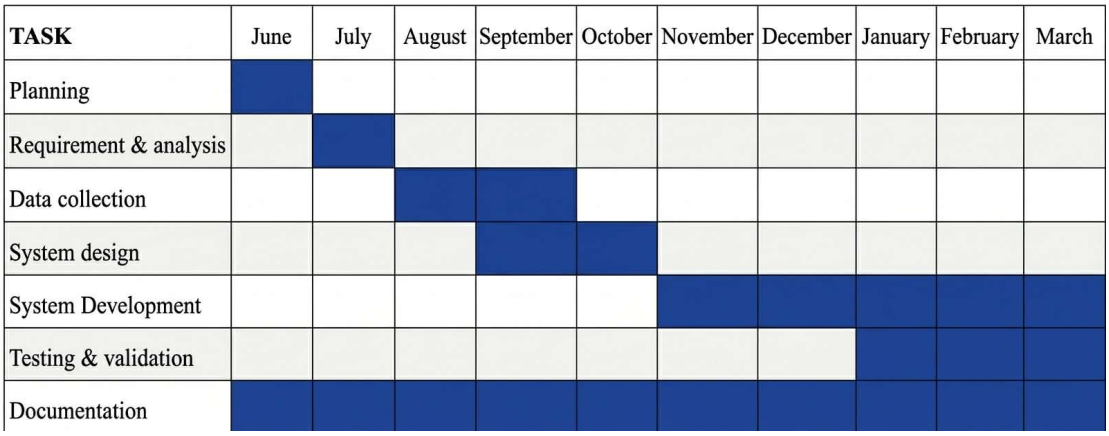

fig. 7.1: *Gantt chart*

CHAPTER 8
CONCLUSION, LIMITATIONS AND FUTURE ENHANCEMENTS

8.1 Conclusion

This project, *AgroLedger: A Blockchain-Based Agricultural Supply Chain Management System*, demonstrates how blockchain technology can be effectively utilized to address the key challenges present in traditional agricultural supply chains, particularly in developing countries like Nepal. Existing systems often suffer from poor transparency, inefficient record-keeping, and a heavy dependence on intermediaries, which leads to reduced trust and unfair pricing for farmers. The proposed system integrates blockchain with a decentralized application (DApp) to create a secure, transparent, and traceable platform for managing agricultural products. By using Ethereum-based smart contracts, the system ensures that all transactions are recorded in an immutable manner, preventing data tampering and enhancing trust among stakeholders. Features such as role-based access, product tracking, and end-to-end traceability allow farmers, distributors, retailers, and consumers to interact within a unified system. Overall, the implementation highlights the potential of blockchain technology to modernize agricultural supply chains by improving transparency, empowering farmers, and enabling consumers to verify product authenticity.

8.2 Limitations

Despite the successful implementation of the AgroLedger system, several limitations exist that affect its real-world applicability. One major limitation is that the system is currently deployed on a local blockchain network using Ganache, which restricts accessibility and does not fully represent real-world scalability challenges. Deployment on public blockchain networks may introduce additional issues such as high transaction costs (gas fees) and network latency, which can impact system performance and usability. Another limitation is the reliance on manual data entry at various stages of the supply chain, which creates the possibility of incorrect or fraudulent data being recorded on the blockchain, as the system itself cannot verify the authenticity of input data. Furthermore, the lack of integration with external technologies such as IoT devices and real-time monitoring systems limits the ability to automate data collection and ensure higher accuracy. In the context of Nepal, a significant practical limitation is the low level of technological literacy among farmers and rural stakeholders, as many users

are not familiar with smartphones, digital platforms, or blockchain-based systems, and may have limited access to reliable internet connectivity, making it difficult for them to effectively interact with the system. Additionally, the user interface, although functional, may not be sufficiently intuitive for non-technical users, which can further hinder adoption, and the system has not yet been tested in large-scale real-world environments, making it difficult to evaluate its performance, scalability, and long-term reliability.

8.3 Future Enhancements

The AgroLedger system can be significantly improved through several future enhancements that aim to increase its practicality, scalability, and usability. One important improvement is deploying the system on a public Ethereum testnet or mainnet, which would allow real-world usage and provide insights into performance under realistic conditions. Integrating IoT devices such as sensors can automate data collection related to environmental conditions like temperature and humidity, thereby improving data accuracy and reducing dependency on manual input. Developing a mobile application can further enhance accessibility, especially for farmers who may rely on smartphones for daily operations, and incorporating multilingual support can increase usability across different user groups. The system can also be extended by integrating external APIs to provide real-time information such as weather updates, market prices, and logistics tracking, making it more useful for decision-making. Additionally, implementing advanced security features such as identity verification and fraud detection mechanisms can improve trust and reliability. Future work can also focus on optimizing the system to reduce transaction costs and improve scalability, ensuring that it can be effectively adopted in developing countries.

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